

QuakeNet — Work Distribution

CSE421 — Computer Network | Section 07 | Group 703

Member	Primary Role	Responsibilities
Tahamidul Alam Chowdhury 22299066	Design & Documentation Lead	Derived the base network address (11.35.0.0/16) and the custom Administrative Distance (144) from the group's member IDs; designed the overall topology, including the warning-loop triangle, the shared central-switch backbone, and the direct and stub links; carried out the complete VLSM subnetting of the address block across all 14 required subnets. Also compiled, wrote, and formatted the final project report — topology figure, VLSM tree diagram, all reference tables, and the assumptions section.
Bornil Barua 22201838	Build & Infrastructure	Built the physical topology in Packet Tracer: placed and correctly named all 43 devices (7 routers, 8 switches, 14 PCs, 7 printers, 7 servers), installed HWIC-2T serial modules on every router, and cabled all 45 connections (39 Ethernet links and 6 serial point-to-point links). Also configured static IP addressing — address, subnet mask, gateway, and DNS — on every router interface, server, and printer according to the addressing plan.
Tabia Sultana Ritu 23201510	Routing & DHCP	Configured DHCP through all three methods used in the design: router-based (IOS) on NEOC for the RIP domain, server-based on MEC and CRC, and relay via ip helper-address on SMS, FRH, USR, and MOP. Set up RIPv2 across the NEOC–SMS–FRH warning loop with static-route redistribution, and implemented the floating and recursive backup static routes (administrative distance 144) on MEC and CRC, including verifying their failover behaviour.
Syed Kafi 23101135	Services & Verification	Configured all application-layer services: email (all seven sites), and web and DNS (NEOC and MEC). Ran the full verification suite — the end-to-end connectivity ping matrix, DHCP lease checks, the MEC floating-route failover demonstration, warning-loop resilience testing, and the cross-domain email and web/DNS checks — and captured the screenshots used as evidence throughout Section 9 of the report.

Note: while each member led one functional area, all four members participated in the design review, the Packet Tracer build, and the end-to-end testing sessions.